

Finite Element Methods

Theory, Geometry, and Computation

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Chapter 1

Finite elements for elasticity

1.1 Introduction and Scope

Elasticity describes the mechanical response of solid bodies subjected to external loads. When forces are applied to a solid, it deforms; when the forces are removed, the solid recovers its original shape. Understanding and predicting this behavior is essential in many fields, including structural engineering, materials science, biomechanics and mechanical design.

In most realistic geometries and loading configurations, analytical solutions of elasticity problems are unavailable. This motivates the use of numerical methods, among which the finite element method (FEM) has become the standard tool. FEM is particularly well suited to elasticity because the governing equations arise naturally from an energy minimization principle, which can be discretized in a systematic and robust way.

This lecture introduces finite element methods for linear elasticity from an energy-based perspective, with the goal of providing both a theoretical foundation and a practical framework for implementation.

Why an energy-based formulation? Elastic behavior is fundamentally governed by the storage and release of mechanical energy. For small deformations, the state of an elastic body is fully characterized by its displacement field, and equilibrium corresponds to a minimum of the total potential energy.

Formulating elasticity in terms of energy has several advantages:

- It provides a direct and physically intuitive description of equilibrium.
- The weak (variational) form emerges naturally through the first variation of the energy.
- Boundary conditions appear in a systematic way: displacement constraints are imposed on the admissible space, while traction conditions arise naturally through work given to the body.
- The resulting variational problem is directly compatible with finite element discretization.

Because of these properties, the finite element method is best understood not as a discretization of partial differential equations, but as a discretization of an energy minimization problem.

Scope of the lecture. This lecture focuses on linear elasticity in the regime of small deformations. The following modeling assumptions are made throughout:

- Deformations are sufficiently small so that linear strain measures are valid.
- The material behavior is elastic, with no plasticity, damage, or hysteresis.

- Inertia effects are neglected; only quasi-static equilibrium is considered. This means that the equilibrium position of a body only depends on the current loading applied to it and not on the history of this loading.
- The body is modeled as a continuous medium (continuum mechanics framework).

Within this setting, the lecture covers both isotropic and anisotropic materials, with particular attention paid to nearly incompressible behavior, which poses specific challenges for finite element discretizations.

Objectives. The main objectives of this lecture are:

1. To present linear elasticity as a variational problem derived from an energy functional.
2. To explain how boundary conditions emerge from the variational formulation.
3. To introduce vector-valued finite element spaces suitable for elasticity.
4. To discuss numerical issues specific to elasticity, such as rigid body modes and incompressibility.

The emphasis is on conceptual clarity and structural understanding, rather than exhaustive mathematical proofs.

What is not covered. To keep the scope focused, several important topics are intentionally left out or only briefly mentioned:

- Nonlinear elasticity and large deformations
- Plasticity and inelastic material behavior
- Dynamic problems and time-dependent effects
- Contact problems
- Detailed error analysis and a priori estimates

These topics require additional modeling tools and are better treated in specialized courses or texts.

1.2 Kinematics of Linear Elasticity

Kinematics describes how a solid body deforms, independently of the forces that cause the deformation. In elasticity, kinematics provides the link between the displacement field and measures of deformation such as strain. This section introduces the kinematic assumptions underlying linear elasticity, with particular emphasis on small deformations and infinitesimal strain.

Displacement field. Let $\Omega \subset \mathbb{R}^d$, $d = 2, 3$, denote the reference (undeformed) configuration of a solid body. The deformation of the body is described by the displacement field

$$\underline{x} = \underline{x}_{\text{deformed}} - \underline{x},$$

which assigns to each material point $x \in \Omega$ a displacement vector.

In linear elasticity, the displacement field is assumed to be small compared to the characteristic dimensions of the body and sufficiently smooth to admit first derivatives. Under these assumptions, all quantities are evaluated with respect to the reference configuration.

Displacement gradient and linearization. The gradient of the displacement field, ∇u , measures local variations of the displacement. It contains information about stretching, shear, and rotation. However, ∇u itself is not an appropriate strain measure, since it does not distinguish between actual deformation and rigid body motion.

Infinitesimal strain tensor. The fundamental strain measure in linear elasticity is the infinitesimal strain tensor, defined as the symmetric part of the displacement gradient:

$$\underline{\underline{\varepsilon}}(\underline{u}) = \frac{1}{2}(\nabla \underline{u} + (\nabla \underline{u})^T).$$

The infinitesimal strain tensor has the following properties:

- it is symmetric and so its eigenvalues are real,
- it depends only on first derivatives of the displacement,
- it measures changes in length and angles to first order,
- it vanishes for rigid body motions.

The diagonal components represent normal strains, while the off-diagonal components represent shear strains.

Rigid body motions. A rigid body motion consists of a translation and a rotation. In the linearized setting, a rigid body displacement can be written as

$$\underline{u}(x) = \underline{a} + \underline{R} \cdot \underline{x},$$

where $\underline{a} \in \mathbb{R}^d$ is a constant translation vector and $\underline{R} \in \mathbb{R}^{d \times d}$ is a skew-symmetric matrix representing an infinitesimal rotation.

For such displacements, the infinitesimal strain vanishes:

$$\underline{\underline{\varepsilon}}(\underline{u}) = \underline{0}.$$

This property ensures that no elastic energy is associated with rigid body motions and explains why the strain tensor, rather than the full displacement gradient, is the relevant kinematic quantity in elasticity.

Volumetric and deviatoric strain. The strain tensor can be decomposed into volumetric and deviatoric parts. The volumetric strain is given by the trace of the strain tensor and measures local volume change. The deviatoric strain represents shape changes at constant volume.

This decomposition plays a central role in elasticity, particularly for nearly incompressible materials, where volumetric deformations are strongly penalized in the elastic energy.

Compatibility of strain. Not every symmetric tensor field corresponds to the strain of a displacement field. A strain field is said to be compatible if there exists a displacement field whose infinitesimal strain it represents. While compatibility conditions are important from a theoretical standpoint, displacement-based finite element formulations automatically enforce compatibility.

Kinematic assumptions of linear elasticity. The kinematic framework of linear elasticity relies on the following assumptions:

- displacements and their gradients are small,
- rotations are small and treated only to first order,
- the infinitesimal strain tensor provides an accurate measure of deformation,
- changes in geometry are neglected and evaluated in the reference configuration.

Summary. The kinematics of linear elasticity establishes the relationship between displacement and strain under the assumption of small deformations. The infinitesimal strain tensor is the central kinematic quantity, capturing all relevant deformation modes while remaining invariant under rigid body motions. This framework forms the foundation for the definition of elastic energy and the variational formulations used in finite element methods.

1.3 Elasticity as an Energy Minimization Problem

In linear elasticity, the equilibrium configuration of a solid body is characterized by the minimization of a scalar energy functional. Rather than taking forces or stresses as primary quantities, we adopt an energy-based viewpoint in which material behavior is described entirely through the energetic cost of deformation. This perspective is physically transparent, mathematically robust, and particularly well suited for finite element discretization.

1.3.1 Stored elastic energy density.

For small deformations, the mechanical energy stored in a material depends on the infinitesimal strain tensor $\underline{\underline{\varepsilon}}$. In linear elasticity, the stored elastic energy density is assumed to be a quadratic function of the strain:

$$W(\underline{\underline{\varepsilon}}) = \frac{1}{2} \underline{\underline{\varepsilon}} : \underline{\underline{C}} : \underline{\underline{\varepsilon}},$$

where $\underline{\underline{C}}$ is a fourth-order elasticity tensor and the symbol “:” denotes double tensor contraction.

This quadratic form reflects three fundamental assumptions: the reference configuration is an energy minimum, the material response is linear, and the energy depends only on strain and not on rigid body motion. The energy density satisfies

$$W(\underline{\underline{0}}) = 0, \quad W(\underline{\underline{\varepsilon}}) > 0 \quad \text{for all } \underline{\underline{\varepsilon}} \neq \underline{\underline{0}},$$

ensuring stability.

1.3.2 Mathematical structure of the energy.

The elasticity tensor $\underline{\underline{C}}$ defines a bilinear form on the space of symmetric tensors. For the energy density to be physically meaningful, $\underline{\underline{C}}$ must satisfy the minor symmetries

$$C_{ijkl} = C_{jikl} = C_{ijlk},$$

the major symmetry

$$C_{ijkl} = C_{klij},$$

and the positive definiteness condition

$$\underline{\underline{\varepsilon}} : \underline{\underline{C}} : \underline{\underline{\varepsilon}} \geq c \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}} \quad \text{for some } c > 0.$$

1.3.3 Total elastic energy of the body.

Let u denote the displacement field and $\underline{\underline{\varepsilon}}(\underline{u})$ the associated infinitesimal strain. The total elastic energy stored in the body is given by

$$E_{\text{int}}(\underline{u}) = \int_{\Omega} \frac{1}{2} \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{u}) \, dx.$$

The displacement enters the formulation only through its strain, and rigid body motions do not contribute to the elastic energy.

1.3.4 External work.

External actions perform work on the body. We consider body forces f acting in Ω and surface forces t acting on $\Gamma_N \subset \partial\Omega$. The corresponding work functionals are

$$W_{\text{body}}(\underline{u}) = \int_{\Omega} \underline{f} \cdot \underline{u} \, dx, \quad W_{\text{surf}}(\underline{u}) = \int_{\Gamma_N} \underline{t} \cdot \underline{u} \, ds.$$

1.3.5 Total potential energy.

The total potential energy of the elastic body is defined as

$$\Pi(\underline{u}) = E_{\text{int}}(\underline{u}) - W_{\text{body}}(\underline{u}) - W_{\text{surf}}(\underline{u}),$$

that is,

$$\Pi(\underline{u}) = \int_{\Omega} \frac{1}{2} \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{u}) \, dx - \int_{\Omega} \underline{f} \cdot \underline{u} \, dx - \int_{\Gamma_N} \underline{t} \cdot \underline{u} \, ds.$$

Mechanical equilibrium corresponds to a displacement field $\underline{u}$ that minimizes $\Pi(\underline{u})$ among all admissible displacements.

1.3.6 Isotropic elastic energy.

A material is isotropic if its elastic response is invariant under arbitrary rotations. In this case, the elasticity tensor depends on only two material parameters λ and μ . The elastic energy density takes the form

$$W(\underline{\underline{\varepsilon}}) = \mu \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}} + \frac{\lambda}{2} (\text{tr } \underline{\underline{\varepsilon}})^2.$$

This expression separates shear and volumetric contributions. Note that if we add a term related to $\det(\underline{\underline{\varepsilon}})$, the energy will still be isotropic but no longer quadratic in the strain. As a consequence, the variation of the energy will yield nonlinear elasticity equation instead of linear equations.

1.3.7 Volumetric–deviatoric decomposition.

The strain tensor admits the decomposition

$$\underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}}_{\text{dev}} + \frac{1}{d} (\text{tr } \underline{\underline{\varepsilon}}) \underline{\underline{I}},$$

where d is the spatial dimension. The isotropic energy can then be written as

$$W(\underline{\underline{\varepsilon}}) = \mu \underline{\underline{\varepsilon}}_{\text{dev}} : \underline{\underline{\varepsilon}}_{\text{dev}} + \frac{\kappa}{2} (\text{tr } \underline{\underline{\varepsilon}})^2, \quad \kappa = \lambda + \frac{2\mu}{d}.$$

1.3.8 Anisotropic elastic energy.

A material is anisotropic if its elastic response depends on direction. In this case, the elasticity tensor $\underline{\underline{C}}$ is not invariant under arbitrary rotations and contains additional independent coefficients. Nevertheless, the energy density retains the general quadratic form

$$W(\underline{\underline{\varepsilon}}) = \frac{1}{2} \underline{\underline{\varepsilon}} : \underline{\underline{C}} : \underline{\underline{\varepsilon}}.$$

1.3.9 Types of anisotropy.

In three dimensions, a fully anisotropic material is characterized by 21 independent elastic coefficients. Material symmetries reduce this number:

- *Orthotropic materials* possess three orthogonal planes of symmetry and are described by 9 independent coefficients.
- *Transversely isotropic materials* are isotropic in a plane and have 5 independent coefficients.
- *Cubic symmetry* leads to 3 independent coefficients and lies between isotropy and full anisotropy.

In all cases, the variational structure of the problem remains unchanged; only the elasticity tensor differs.

1.3.10 Plane Strain vs Plane Stress

Many elasticity problems are effectively two-dimensional, either because the geometry is thin in one direction or because deformations are constrained. In such situations, a three-dimensional elastic model is reduced to a two-dimensional one. Two classical reductions are plane strain and plane stress. Although both lead to two-dimensional formulations, they rely on different physical assumptions and result in different elastic energy densities.

Plane strain. Plane strain applies to bodies that are very long or constrained in the out-of-plane direction. Typical examples include long tunnels, dams, or thick structures whose deformation in the third direction is negligible.

In plane strain, the displacement has no out-of-plane component, and all displacement components are independent of the out-of-plane coordinate. As a consequence, the out-of-plane strain components vanish, while the in-plane strains remain free.

From the energy perspective, plane strain corresponds to restricting the three-dimensional elastic energy to displacement fields satisfying these kinematic constraints. No energetic relaxation is performed: the material behaves locally as a three-dimensional elastic solid.

For an isotropic material with Lamé parameters λ and μ , the two-dimensional plane strain elastic energy density (per unit thickness), expressed in terms of the in-plane strain tensor $\underline{\underline{\varepsilon}}$, is

$$W_{\text{psn}}(\underline{\underline{\varepsilon}}) = \mu \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}} + \frac{\lambda}{2} (\text{tr } \underline{\underline{\varepsilon}})^2.$$

Because out-of-plane deformation is suppressed, plane strain models are generally stiffer.

Plane stress. Plane stress applies to thin structures such as plates or sheets that are free to deform in the thickness direction and carry negligible stress across the thickness.

In plane stress, the out-of-plane stress components are assumed to vanish. The out-of-plane strain is not constrained and adjusts itself so as to minimize the elastic energy under this condition.

From the energy viewpoint, plane stress is obtained by minimizing the three-dimensional elastic energy with respect to the out-of-plane strain components, under the assumption that no out-of-plane stresses are present. This leads to an effective two-dimensional energy density depending only on the in-plane strains.

For an isotropic material, the resulting two-dimensional plane stress elastic energy density is

$$W_{\text{pst}}(\underline{\underline{\varepsilon}}) = \mu \underline{\underline{\varepsilon}} : \underline{\underline{\varepsilon}} + \frac{\lambda\mu}{\lambda + 2\mu} (\text{tr } \underline{\underline{\varepsilon}})^2.$$

Compared to plane strain, the volumetric contribution is reduced, reflecting the fact that the material can deform freely in the thickness direction.

Comparison from an energy perspective. Although plane strain and plane stress both lead to quadratic energy densities in terms of the in-plane strain tensor, the coefficients multiplying the volumetric term differ:

- in plane strain, the volumetric term is proportional to λ ,
- in plane stress, the volumetric term is proportional to $\lambda\mu/(\lambda + 2\mu)$.

Since $\lambda\mu/(\lambda + 2\mu) < \lambda$, plane stress always leads to a more compliant response than plane strain for the same material parameters.

When to use each model. Plane strain should be used for thick or constrained structures where out-of-plane deformation is prevented. Plane stress should be used for thin structures where out-of-plane stresses are negligible. Using an inappropriate model leads to incorrect energy storage and inaccurate predictions of stiffness and deformation.

Summary. Plane strain and plane stress are two distinct two-dimensional reductions of three-dimensional elasticity. From an energy viewpoint, plane strain enforces kinematic constraints, while plane stress results from energetic relaxation. Their elastic energy densities differ accordingly, leading to different mechanical responses even for the same material.

Nearly incompressible and incompressible behavior. Nearly incompressible materials strongly penalize volume changes, corresponding to a bulk modulus κ much larger than the shear modulus μ . In the incompressible limit, admissible deformations satisfy the constraint

$$\text{tr } \underline{\underline{\varepsilon}}(\underline{u}) = 0.$$

From the energy viewpoint, incompressibility appears as a constraint or as a penalty on volumetric strain, a fact that has important numerical consequences.

Equilibrium as an energy minimum. The equilibrium displacement minimizes the total potential energy $\Pi(\underline{u})$. This minimization principle is equivalent to the vanishing of the first variation of Π and forms the basis for the variational and finite element formulations developed in subsequent sections.

1.3.11 Summary.

Linear elasticity can be formulated entirely in terms of energy. The elastic response of a material is encoded in a quadratic energy density depending on the infinitesimal strain, while external actions enter through linear work terms. This energy-based viewpoint provides a unified framework for isotropic and anisotropic materials and constitutes the natural starting point for finite element methods in elasticity.

1.4 Variational Formulation and Boundary Conditions

Let $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) be the reference configuration of the body, with boundary decomposition

$$\partial\Omega = \Gamma_D \cup \Gamma_N, \quad \Gamma_D \cap \Gamma_N = \emptyset.$$

The unknown is the displacement field

$$\underline{u} : \Omega \rightarrow \mathbb{R}^d.$$

Strain and elastic energy. The infinitesimal strain tensor is defined as

$$\underline{\underline{\varepsilon}}(\underline{u}) = \frac{1}{2}(\nabla \underline{u} + \nabla \underline{u}^T),$$

and the elastic energy density is

$$W(\underline{\underline{\varepsilon}}) = \frac{1}{2} \underline{\underline{\varepsilon}} : \underline{\underline{C}} : \underline{\underline{\varepsilon}},$$

where $\underline{\underline{C}}$ is the fourth-order elasticity tensor.

Admissible spaces. We introduce the admissible displacement space

$$\mathcal{U} = \{\underline{u} \in H^1(\Omega)^d : \underline{u} = \underline{u}_D \text{ on } \Gamma_D\},$$

and the space of virtual displacements

$$\mathcal{V} = \{\underline{v} \in H^1(\Omega)^d : \underline{v} = \underline{0} \text{ on } \Gamma_D\}.$$

Total potential energy. The total potential energy functional is defined by

$$\Pi(\underline{u}) = \int_{\Omega} W(\underline{\underline{\varepsilon}}(\underline{u})) dx - \int_{\Omega} \underline{f} \cdot \underline{u} dx - \int_{\Gamma_N} \underline{t} \cdot \underline{u} ds,$$

where $\underline{f}$ denotes the body force and $\underline{t}$ boundary forces (also called tractions acting on the boundary).

Principle of minimum potential energy. The equilibrium displacement $\underline{u} \in \mathcal{U}$ satisfies

$$\underline{u} = \arg \min_{\underline{w} \in \mathcal{U}} \Pi(\underline{w}).$$

Equivalently, $\underline{u}$ is characterized by the stationarity condition

$$\delta \Pi(\underline{u}; \underline{v}) = 0 \quad \forall \underline{v} \in \mathcal{V}.$$

First variation. Let $u_{\eta} = \underline{u} + \eta v$, with $v \in \mathcal{V}$. The first variation of the energy reads

$$\delta \Pi(\underline{u}; \underline{v}) = \int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v}) dx - \int_{\Omega} \underline{f} \cdot \underline{v} dx - \int_{\Gamma_N} \underline{t} \cdot \underline{v} ds.$$

where

$$\underline{\underline{\sigma}}(\underline{u}) = \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{u})$$

The second-order tensor $\underline{\underline{\sigma}}$ is called the Cauchy stress. Note that we have obtained the stress quantity by taking the variation of the energy. The stress is the quantity acting on the test strain $\underline{\underline{\varepsilon}}(\underline{v})$.

If you are not convinced of the above, here are some details :

$$W(\underline{\underline{\varepsilon}}(\underline{u} + \eta \underline{v})) = \frac{1}{2}(\underline{\underline{\varepsilon}}(\underline{u}) + \eta \underline{\underline{\varepsilon}}(\underline{v})) : \underline{\underline{C}} : (\underline{\underline{\varepsilon}}(\underline{u}) + \eta \underline{\underline{\varepsilon}}(\underline{v})) \quad (1.1)$$

$$= \frac{1}{2} \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{u}) + \underbrace{\eta \underline{\underline{\varepsilon}}(\underline{u}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{v})}_{\underline{\underline{\sigma}}} + \frac{1}{2} \eta^2 \underline{\underline{\varepsilon}}(\underline{v}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{v}) \quad (1.2)$$

Weak formulation. The stationarity condition therefore becomes

$$\int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v}) \, dx - \int_{\Omega} \underline{f} \cdot \underline{v} \, dx - \int_{\Gamma_N} \underline{t} \cdot \underline{v} \, ds = 0 \quad \forall \underline{v} \in \mathcal{V}.$$

The weak formulation of linear elasticity reads: find $\underline{u} \in \mathcal{U}$ such that

$$a(\underline{u}, \underline{v}) = \ell(\underline{v}) \quad \forall \underline{v} \in \mathcal{V},$$

with bilinear form

$$a(\underline{u}, \underline{v}) = \int_{\Omega} \underline{\underline{\varepsilon}}(\underline{v}) : \underline{\underline{C}} : \underline{\underline{\varepsilon}}(\underline{u}) \, dx,$$

and linear functional

$$\ell(\underline{v}) = \int_{\Omega} \underline{f} \cdot \underline{v} \, dx + \int_{\Gamma_N} \underline{t} \cdot \underline{v} \, ds.$$

Strong form and emergence of boundary conditions. Using integration by parts,

$$\int_{\Omega} \underline{\underline{\sigma}}(\underline{u}) : \underline{\underline{\varepsilon}}(\underline{v}) \, dx = - \int_{\Omega} (\nabla \cdot \underline{\underline{\sigma}}(\underline{u})) \cdot \underline{v} \, dx + \int_{\partial\Omega} (\underline{\underline{\sigma}}(\underline{u}) \cdot \underline{n}) \cdot \underline{v} \, ds.$$

Since $\underline{v} = 0$ on Γ_D , boundary terms only arise on Γ_N , we thus have

$$\int_{\Omega} (\nabla \cdot \underline{\underline{\sigma}}(\underline{u}) + \underline{f}) \cdot \underline{v} \, dx - \int_{\Gamma_N} (\underline{\underline{\sigma}}(\underline{u}) \cdot \underline{n} - \underline{t}) \cdot \underline{v} \, ds = 0 \quad \forall \underline{v} \in \mathcal{V}.$$

If $\underline{u}$ is sufficiently smooth, the weak formulation implies the so-called strong form

$$\begin{cases} -\nabla \cdot \underline{\underline{\sigma}}(\underline{u}) = \underline{f} & \text{in } \Omega, \\ \underline{u} = \underline{u}_D & \text{on } \Gamma_D, \\ \underline{\underline{\sigma}}(\underline{u}) \cdot \underline{n} = \underline{t} & \text{on } \Gamma_N. \end{cases}$$

The imposed tractions must thus correspond to the normal stress on the boundary.

Well-posedness. The bilinear form $a(\cdot, \cdot)$ is symmetric and continuous. Coercivity on $\mathcal{V}$ follows from Korn's inequality,

$$\|\underline{u}\|_{H^1(\Omega)} \leq C \|\underline{\underline{\varepsilon}}(\underline{u})\|_{L^2(\Omega)} \quad \forall \underline{u} \in \mathcal{V}.$$

If the Dirichlet boundary Γ_D is insufficient, rigid body motions belong to the kernel of the bilinear form.

Stress-strain relations We have observed the stress appearing when taking the first variation of the energy. A more general definition is

$$\underline{\underline{\sigma}} = \frac{\partial W(\underline{\underline{\varepsilon}})}{\partial \underline{\underline{\varepsilon}}}$$

which is valid even if W is not quadratic in the strain. The stress strain relation is

$$\underline{\underline{\sigma}} = \underline{\underline{C}} : \underline{\underline{\varepsilon}} \quad (\text{fully anisotropic}) \tag{1.3}$$

$$\underline{\underline{\sigma}} = 2\mu \underline{\underline{\varepsilon}} + \lambda(\text{tr } \underline{\underline{\varepsilon}}) \underline{\underline{I}} \quad (\text{isotropic 3D}) \tag{1.4}$$

$$\underline{\underline{\sigma}} = 2\mu \underline{\underline{\varepsilon}} + \lambda(\text{tr } \underline{\underline{\varepsilon}}) \underline{\underline{I}} \quad (\text{isotropic 2D plane strain}) \tag{1.5}$$

$$\underline{\underline{\sigma}} = 2\mu \underline{\underline{\varepsilon}} + \frac{2\lambda\mu}{\lambda + 2\mu} (\text{tr } \underline{\underline{\varepsilon}}) \underline{\underline{I}} \quad (\text{isotropic 2D plane stress}) \tag{1.6}$$

1.5 Vectorial Finite Element Approximation

In linear elasticity, the finite element method approximates a vector-valued displacement field by restricting the energy minimization problem to a finite-dimensional space. Because the unknown has several components, the discretization must correctly represent how these components are coupled through the elastic energy. This section explains how vector-valued finite element spaces are constructed, how the discrete system is obtained, and how elementary stiffness matrices are computed and assembled in practice.

1.5.1 Vector-Valued Finite Element Spaces

The displacement field has one component per spatial direction. In most implementations, vector-valued finite element spaces are constructed component-wise from a scalar finite element space.

The standard procedure is to choose a scalar finite element space (for example, continuous piecewise linear or quadratic functions) and to approximate each displacement component using the same scalar space. As a result, each mesh node carries multiple degrees of freedom: two in two dimensions and three in three dimensions. This construction is simple, flexible, and supported by most finite element libraries.

1.5.2 Discrete Admissible and Test Spaces

The continuous admissible space of displacements is replaced by a finite-dimensional subspace that satisfies the prescribed displacement boundary conditions. Degrees of freedom associated with Dirichlet boundaries are constrained.

The test space consists of the same finite element functions but vanishing on the Dirichlet boundary. In practice, this distinction is handled by modifying the global system to account for constrained degrees of freedom.

1.5.3 Discrete Variational Problem

Restricting the total potential energy to the finite element space leads to a finite-dimensional minimization problem. Equivalently, the discrete problem consists in finding a finite element displacement field such that the discrete weak form holds for all finite element test functions.

This results in a linear system of the form

$$\text{global stiffness matrix} \times \text{displacement vector} = \text{load vector}.$$

The stiffness matrix is symmetric and sparse. It is positive definite when rigid body modes are eliminated by boundary conditions.

1.5.4 Basis Functions and Degrees of Freedom

Let $\{\phi_i\}$ denote the scalar finite element shape functions. Each scalar shape function gives rise to several vector-valued basis functions, one for each displacement component.

In two dimensions, the vector-valued basis functions associated with ϕ_i are

$$\phi_i \underline{e}_1 = (\phi_i, 0), \quad \phi_i \underline{e}_2 = (0, \phi_i),$$

where $\underline{e}_1$ and $\underline{e}_2$ are the Cartesian unit vectors. This component-wise construction leads to a natural block structure in the stiffness matrix, reflecting the coupling between displacement components induced by the elastic energy.

1.5.5 Element-Level Elastic Energy

On an element K , the elastic energy associated with a displacement field u is

$$E_K(\underline{u}) = \int_K \frac{1}{2} \underline{\varepsilon}(\underline{u}) : \underline{\underline{C}} : \underline{\varepsilon}(\underline{u}) \, dx.$$

When u is approximated by a finite element expansion, this energy becomes a quadratic form in the element degrees of freedom. The coefficients of this quadratic form define the elementary stiffness matrix.

1.5.6 Elementary Stiffness Matrix: Abstract Form

Let

$$\{\Phi_a\}_{a=1}^{N_{\text{dof}}^K}$$

denote the vector-valued basis functions associated with an element K , where each index a corresponds to a single displacement degree of freedom (one component at one node).

The entries of the element stiffness matrix are defined by

$$K_{ab}^{(K)} = \int_K \underline{\varepsilon}(\Phi_b) : \underline{\underline{C}} : \underline{\varepsilon}(\Phi_a) \, dx.$$

The matrix $K^{(K)}$ has size $N_{\text{dof}}^K \times N_{\text{dof}}^K$ and is symmetric by construction.

1.5.7 Component-Wise Construction and Index Structure

We now make explicit the component-wise construction of the vector-valued basis functions. Let

$$\{\phi_i\}_{i=1}^{N_{\text{shape}}^K}$$

be the scalar shape functions on element K , and let $\underline{e}_\alpha$, $\alpha = 1, \dots, d$, denote the Cartesian unit vectors.

Each vector-valued basis function is defined as

$$\Phi_{i,\alpha} = \phi_i \underline{e}_\alpha.$$

Here, i labels the scalar shape function, α labels the displacement component, and the combined index

$$a = (i, \alpha)$$

labels a vector degree of freedom.

With this notation, the element stiffness matrix entries take the form

$$K_{(i,\alpha),(j,\beta)}^{(K)} = \int_K \underline{\varepsilon}(\phi_j \underline{e}_\beta) : \underline{\underline{C}} : \underline{\varepsilon}(\phi_i \underline{e}_\alpha) \, dx.$$

This expression makes explicit the coupling between scalar shape functions and displacement components through the elastic energy.

1.5.8 Matrix Form for Implementation

For implementation purposes, it is convenient to express the element stiffness matrix in matrix form. Introducing the strain–displacement matrix B , the element stiffness matrix can be written as

$$K^{(K)} = \int_K B^T C B \, dx,$$

where C is the matrix representation of the elasticity tensor. Numerical quadrature is used to evaluate this integral.

1.5.9 Assembly into the Global System

Each element stiffness matrix is assembled into the global stiffness matrix using the mesh connectivity. Local degrees of freedom a are mapped to global indices, and element contributions are accumulated.

Because each element couples only nearby degrees of freedom, the global stiffness matrix is sparse. Efficient sparse storage and assembly are essential for large-scale elasticity computations.

1.5.10 Rigid Body Modes

Rigid body motions do not produce strain and therefore do not contribute to the elastic energy. If displacement boundary conditions are insufficient, the global stiffness matrix has a nontrivial nullspace.

In practice, rigid body modes are eliminated by fixing selected displacement degrees of freedom, adding constraints, or explicitly handling the nullspace in iterative solvers.

1.5.11 Boundary Conditions at the Discrete Level

Displacement boundary conditions are typically imposed strongly by eliminating or constraining degrees of freedom.

Traction boundary conditions contribute only to the load vector. At the element level, boundary integrals are computed on faces or edges and assembled into the global right-hand side.

1.5.12 Nearly Incompressible Materials

For nearly incompressible materials, the volumetric part of the elastic energy dominates. Standard displacement-based finite element spaces may then suffer from volumetric locking, leading to overly stiff numerical solutions.

This issue requires modified formulations, such as mixed methods or stabilization techniques, which are discussed later.

1.5.13 Summary

Vectorial finite element approximation of elasticity consists in constructing component-wise vector-valued finite element spaces, computing element stiffness matrices from the elastic energy, assembling sparse symmetric global systems, carefully imposing boundary conditions, and handling rigid body modes and incompressibility effects.

The clarified index structure

$$i \text{ (shape function), } \alpha \text{ (component), } a = (i, \alpha) \text{ (degree of freedom)}$$

provides a clean bridge between the mathematical formulation and practical implementation.

1.5.14 Worked 2D P_1 Example (Isotropic Elasticity): Explicit Element Stiffness Matrix

We consider a single linear triangular element K in 2D with vertices

$$(x_1, y_1), \quad (x_2, y_2), \quad (x_3, y_3),$$

and nodal displacement unknowns ordered as

$$u^K = [u_1 \quad v_1 \quad u_2 \quad v_2 \quad u_3 \quad v_3]^T.$$

We assume isotropic linear elasticity in *plane strain* (for definiteness) and introduce a thickness $t > 0$ (often $t = 1$ in pure 2D models).

Shape functions and constant gradients

For P_1 triangles, the scalar shape functions are affine on K :

$$\phi_i(x, y) = \frac{a_i + b_i x + c_i y}{2A}, \quad i = 1, 2, 3,$$

where A is the area of the triangle,

$$A = \frac{1}{2} \det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix},$$

and the coefficients are defined by

$$\begin{aligned} b_1 &= y_2 - y_3, & c_1 &= x_3 - x_2, \\ b_2 &= y_3 - y_1, & c_2 &= x_1 - x_3, \\ b_3 &= y_1 - y_2, & c_3 &= x_2 - x_1. \end{aligned}$$

(Explicit expressions for a_i are not needed here.) The gradients are constant on K :

$$\nabla \phi_i = \frac{1}{2A} \begin{bmatrix} b_i \\ c_i \end{bmatrix}, \quad i = 1, 2, 3.$$

Discrete displacement on the element

Using the scalar basis functions, the displacement approximation on K is

$$u_h(x, y) = \sum_{i=1}^3 u_i \phi_i(x, y), \quad v_h(x, y) = \sum_{i=1}^3 v_i \phi_i(x, y).$$

The infinitesimal strain in 2D is

$$\underline{\underline{\varepsilon}}(\underline{u}_h) = \begin{bmatrix} \partial_x u_h \\ \partial_y v_h \\ \partial_y u_h + \partial_x v_h \end{bmatrix} \quad (\text{Voigt form } [\underline{\varepsilon}_{xx}, \underline{\varepsilon}_{yy}, 2\underline{\varepsilon}_{xy}]^T).$$

Strain–displacement matrix B

For a P_1 triangle, B is constant on K and satisfies

$$\underline{\underline{\varepsilon}}(\underline{u}_h) = B \mathbf{u}^K.$$

With the above ordering of $\mathbf{u}^K$, one obtains

$$B = \frac{1}{2A} \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}.$$

Isotropic constitutive matrix C (plane strain)

In plane strain, the isotropic elasticity matrix in terms of Lamé parameters (λ, μ) is

$$C = \begin{bmatrix} \lambda + 2\mu & \lambda & 0 \\ \lambda & \lambda + 2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}.$$

(If desired, one may use $\mu = \frac{E}{2(1+\nu)}$ and $\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}$.)

Element stiffness matrix: compact formula

Since B and C are constant on K , the element stiffness matrix is

$$K^{(K)} = \int_K t B^T C B \, dx = t A B^T C B.$$

Element stiffness matrix: explicit 6×6 expression

Expanding $K^{(K)} = [K_{mn}^{(K)}]_{m,n=1}^6$ in terms of (b_i, c_i) yields

$$K^{(K)} = \frac{t}{4A} \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{12} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{13} & k_{23} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{14} & k_{24} & k_{34} & k_{44} & k_{45} & k_{46} \\ k_{15} & k_{25} & k_{35} & k_{45} & k_{55} & k_{56} \\ k_{16} & k_{26} & k_{36} & k_{46} & k_{56} & k_{66} \end{bmatrix},$$

with

$$\begin{aligned} k_{11} &= (\lambda + 2\mu)b_1^2 + \mu c_1^2, & k_{12} &= (\lambda + \mu)b_1 c_1, \\ k_{22} &= \mu b_1^2 + (\lambda + 2\mu)c_1^2, \\ k_{13} &= (\lambda + 2\mu)b_1 b_2 + \mu c_1 c_2, & k_{14} &= \lambda b_1 c_2 + \mu b_2 c_1, \\ k_{23} &= \mu b_1 c_2 + \lambda b_2 c_1, & k_{24} &= \mu b_1 b_2 + (\lambda + 2\mu)c_1 c_2, \\ k_{15} &= (\lambda + 2\mu)b_1 b_3 + \mu c_1 c_3, & k_{16} &= \lambda b_1 c_3 + \mu b_3 c_1, \\ k_{25} &= \mu b_1 c_3 + \lambda b_3 c_1, & k_{26} &= \mu b_1 b_3 + (\lambda + 2\mu)c_1 c_3, \\ k_{33} &= (\lambda + 2\mu)b_2^2 + \mu c_2^2, & k_{34} &= (\lambda + \mu)b_2 c_2, \\ k_{44} &= \mu b_2^2 + (\lambda + 2\mu)c_2^2, \\ k_{35} &= (\lambda + 2\mu)b_2 b_3 + \mu c_2 c_3, & k_{36} &= \lambda b_2 c_3 + \mu b_3 c_2, \\ k_{45} &= \mu b_2 c_3 + \lambda b_3 c_2, & k_{46} &= \mu b_2 b_3 + (\lambda + 2\mu)c_2 c_3, \\ k_{55} &= (\lambda + 2\mu)b_3^2 + \mu c_3^2, & k_{56} &= (\lambda + \mu)b_3 c_3, \\ k_{66} &= \mu b_3^2 + (\lambda + 2\mu)c_3^2. \end{aligned}$$

This is an explicit closed-form P_1 triangle stiffness matrix for isotropic elasticity in plane strain.

Numerical example (right triangle)

Consider the right triangle with nodes

$$(x_1, y_1) = (0, 0), \quad (x_2, y_2) = (1, 0), \quad (x_3, y_3) = (0, 1).$$

Then $A = \frac{1}{2}$ and

$$(b_1, b_2, b_3) = (-1, 1, 0), \quad (c_1, c_2, c_3) = (-1, 0, 1).$$

For $t = 1$, $E = 1$, $\nu = 0.3$ (plane strain), i.e.

$$\mu = \frac{E}{2(1 + \nu)}, \quad \lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)},$$

the element stiffness matrix evaluates to

$$K^{(K)} = \begin{bmatrix} \frac{45}{52} & \frac{25}{52} & -\frac{35}{52} & -\frac{5}{26} & -\frac{5}{26} & -\frac{15}{52} \\ \frac{25}{52} & \frac{45}{52} & -\frac{15}{52} & -\frac{5}{26} & -\frac{5}{26} & -\frac{35}{52} \\ -\frac{35}{52} & -\frac{15}{52} & \frac{35}{52} & 0 & 0 & \frac{15}{52} \\ -\frac{5}{26} & -\frac{5}{26} & 0 & \frac{5}{26} & \frac{5}{26} & 0 \\ -\frac{5}{26} & -\frac{5}{26} & 0 & \frac{5}{26} & \frac{5}{26} & 0 \\ -\frac{15}{52} & -\frac{35}{52} & \frac{15}{52} & 0 & 0 & \frac{35}{52} \end{bmatrix}.$$

1.6 Boundary Conditions in Practice

Boundary conditions are a central ingredient of elasticity problems. They describe how a solid body interacts with its environment and are essential for ensuring physical realism, mathematical well-posedness, and numerical stability. In finite element formulations derived from an energy minimization principle, boundary conditions determine both the admissible displacement space and the external work acting on the system.

This section discusses the types of boundary conditions encountered in elasticity, their interpretation from an energy viewpoint, common pitfalls in practice, and typical modeling situations illustrated by standard examples.

1.6.1 Types of Boundary Conditions in Elasticity

In elasticity, boundary conditions describe how a solid body interacts mechanically with its surroundings. From both a physical and variational point of view, boundary conditions determine how the displacement field is constrained and how external actions contribute to the total potential energy.

Three main types of boundary conditions are commonly used in practice:

- displacement (Dirichlet) boundary conditions,
- traction (Neumann) boundary conditions,
- Robin (elastic or Winkler-type) boundary conditions.

Each type plays a distinct role in the energy formulation and in finite element implementations.

Displacement (Dirichlet) boundary conditions. Displacement boundary conditions prescribe the value of the displacement field, or some of its components, on a portion of the boundary. Physically, they model supports, clamps, or imposed motions.

From the variational viewpoint, Dirichlet conditions restrict the admissible displacement space. The total potential energy is minimized over displacement fields that satisfy these constraints. In finite element practice, they are imposed strongly by constraining or eliminating the corresponding degrees of freedom.

Traction (Neumann) boundary conditions. Traction boundary conditions prescribe surface forces acting on the boundary, such as applied loads or pressures. They do not restrict the admissible displacement space.

Instead, Neumann conditions enter the formulation through the work of external forces and appear as linear terms in the total potential energy. In finite element implementations, they contribute only to the load vector and do not modify the stiffness matrix.

Robin (Winkler-type) boundary conditions. Robin boundary conditions represent an intermediate case between displacement and traction conditions. They relate boundary tractions to boundary displacements through a linear elastic law.

A typical example is a Winkler foundation, where the boundary is supported by distributed springs and the traction is proportional to the displacement, often in the normal direction.

From the energy viewpoint, Robin boundary conditions contribute an additional quadratic boundary energy term. They do not restrict the admissible displacement space but increase the stiffness of the system through boundary contributions. In finite element implementations, they are treated by assembling additional boundary stiffness matrices.

1.6.2 Essential and Natural Boundary Conditions

From the variational perspective, boundary conditions are classified as:

- *essential boundary conditions*, which must be satisfied by all admissible displacement fields,
- *natural boundary conditions*, which arise naturally from the stationarity of the energy functional.

Displacement boundary conditions are essential, while traction boundary conditions are natural. Robin boundary conditions occupy an intermediate position, contributing directly to the energy functional through quadratic boundary terms.

1.6.3 Pure Traction Problems and Rigid Body Modes

If no displacement boundary conditions are imposed, the elastic energy is invariant under rigid body motions. In such cases, the minimization problem does not admit a unique solution.

Pure traction problems therefore require special care. Although equilibrium may exist, the displacement field is determined only up to a rigid body motion. In practice, uniqueness is restored by fixing a minimal number of displacement degrees of freedom, imposing additional constraints, or explicitly removing rigid body modes.

1.6.4 Common Pitfalls in Practice

Over-constraining the displacement field. A frequent modeling error is to impose too many displacement constraints. Over-constraining occurs when boundary conditions restrict not only rigid body motions but also physically admissible deformations.

This leads to an artificially stiff response and may significantly distort displacement fields, stored energy, and stresses.

Spurious stresses and artificial constraint effects. Improper boundary conditions often generate spurious stresses, that is, stresses arising purely from artificial constraints rather than physical loading.

Common sources include fully clamped boundaries where only partial constraints are appropriate, incompatible displacement conditions at corners or edges, and abrupt transitions between constrained and unconstrained boundary segments.

1.6.5 Directional Boundary Conditions and Mixed Constraints

In many practical problems, boundary conditions act differently on different displacement components at the same boundary location. Some displacement components may be prescribed, while others are left free or subject to tractions.

From the variational viewpoint, prescribed components restrict the admissible displacement space, while unconstrained components are governed by the balance between elastic energy and

external work. In finite element implementations, this is handled by constraining only selected degrees of freedom at boundary nodes.

1.6.6 Illustrative Examples

Roller support. A roller support prevents motion in one direction while allowing motion in the orthogonal direction. For example, vertical displacement is fixed while horizontal displacement is free. Energetically, only one displacement component is constrained. Fully fixing both components would over-constrain the system and introduce spurious stresses.

Symmetry boundary. On a plane of symmetry, the displacement normal to the plane is zero, while tangential displacement is unconstrained. This enforces mirror symmetry of the deformation without artificially constraining tangential motion and is naturally interpreted as a restriction of the admissible displacement space.

Frictionless contact (simplified model). A frictionless contact with a rigid surface can be modeled by prescribing zero normal displacement while leaving tangential displacement free and imposing zero tangential traction. This corresponds to a Dirichlet condition in the normal direction and a Neumann condition in the tangential direction. Energetically, penetration is prevented while sliding is allowed without resistance.

1.6.7 Boundary Conditions and Energy Consistency

From an energy viewpoint, displacement boundary conditions restrict the admissible displacement space, traction boundary conditions contribute linear work terms, and Robin boundary conditions contribute quadratic boundary energy terms.

Ensuring that boundary conditions are implemented consistently with the energy formulation is essential for physical correctness, symmetry of the discrete system, and numerical stability.

1.6.8 Summary

Boundary conditions are a crucial modeling choice in elasticity. Displacement boundary conditions must be sufficient to eliminate rigid body modes without over-constraining the system, while traction and Robin boundary conditions represent external loading and elastic support.

Directional and mixed boundary conditions are common in practice and arise naturally in applications such as supports, symmetry planes, contact, and foundations. Understanding their physical meaning and their role in the energy formulation is essential for constructing reliable finite element models of elastic solids.

1.7 Incompressibility and Mixed Finite Elements

Incompressibility is a major source of difficulty in the finite element approximation of elasticity. While the continuous elastic model naturally accommodates incompressibility, standard displacement-based finite element methods often fail in this regime. This section explains incompressibility from an energy viewpoint, introduces mixed displacement–pressure formulations, and discusses the mathematical conditions required for stable discretization.

1.7.1 Incompressibility from an Energy Viewpoint

For an isotropic linear elastic material, the elastic energy density can be decomposed as

$$W(\underline{\underline{\varepsilon}}) = \mu \underline{\underline{\varepsilon}}_{\text{dev}} : \underline{\underline{\varepsilon}}_{\text{dev}} + \frac{\kappa}{2} (\text{tr } \underline{\underline{\varepsilon}})^2,$$

where

$$\underline{\underline{\varepsilon}}_{\text{dev}} = \underline{\underline{\varepsilon}} - \frac{1}{d}(\text{tr } \underline{\underline{\varepsilon}})\underline{\underline{I}},$$

μ is the shear modulus and κ is the bulk modulus.

Incompressibility corresponds to the constraint

$$\text{tr } \underline{\underline{\varepsilon}}(\underline{\underline{u}}) = 0 \quad \text{in } \Omega,$$

that is, volume-preserving deformations. In the incompressible limit, $\kappa \rightarrow \infty$, the volumetric term dominates the energy.

1.7.2 Incompressibility as a Constraint

Rather than enforcing incompressibility through an infinite volumetric penalty, it is more natural to treat it as a constraint. The elastic problem can then be written as

$$\min_u \int_{\Omega} \mu \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{u}}) : \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{u}}) dx \quad \text{subject to} \quad \text{tr } \underline{\underline{\varepsilon}}(\underline{\underline{u}}) = 0.$$

This is a constrained minimization problem and naturally leads to the introduction of a Lagrange multiplier.

1.7.3 Mixed Displacement–Pressure Formulation

Introducing a scalar Lagrange multiplier p , called pressure, the mixed formulation reads:

Find (u, p) such that

$$\begin{aligned} \int_{\Omega} 2\mu \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{u}}) : \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{v}}) dx - \int_{\Omega} p \text{tr } \underline{\underline{\varepsilon}}(\underline{\underline{v}}) dx &= \ell(\underline{\underline{v}}), \\ \int_{\Omega} q \text{tr } \underline{\underline{\varepsilon}}(\underline{\underline{u}}) dx &= 0, \end{aligned}$$

for all test functions (v, q) .

The pressure p enforces incompressibility and should be interpreted as a Lagrange multiplier, not as a primary stress variable.

1.7.4 Saddle-Point Structure

The mixed problem has the abstract form

$$\begin{aligned} a(\underline{\underline{u}}, \underline{\underline{v}}) + b(\underline{\underline{v}}, p) &= \ell(\underline{\underline{v}}), \\ b(\underline{\underline{u}}, q) &= 0, \end{aligned}$$

with

$$a(\underline{\underline{u}}, \underline{\underline{v}}) = \int_{\Omega} 2\mu \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{u}}) : \underline{\underline{\varepsilon}}_{\text{dev}}(\underline{\underline{v}}) dx, \quad b(\underline{\underline{v}}, p) = - \int_{\Omega} p \text{tr } \underline{\underline{\varepsilon}}(\underline{\underline{v}}) dx.$$

The corresponding discrete linear system has the block structure

$$\begin{bmatrix} A & B^T \\ B & 0 \end{bmatrix} \begin{bmatrix} U \\ P \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix},$$

which is symmetric but indefinite. This is the hallmark of a saddle-point problem.

1.7.5 Inf–Sup (LBB) Condition

Stability of the mixed formulation is governed by the inf–sup (Ladyzhenskaya–Babuška–Brezzi) condition:

$$\inf_{p_h \in Q_h} \sup_{\underline{v}_h \in V_h} \frac{b(\underline{v}_h, p_h)}{\|\underline{v}_h\|_V \|p_h\|_Q} \geq \beta > 0,$$

where β is independent of the mesh size.

This condition ensures that:

- the pressure field is uniquely determined,
- incompressibility is enforced correctly,
- no spurious pressure modes appear.

1.7.6 Meaning of the Inf–Sup Condition

Intuitively, the inf–sup condition expresses a compatibility requirement between the displacement space V_h and the pressure space Q_h . The displacement space must be rich enough to interact meaningfully with all pressure modes.

If the condition is violated, the discrete solution may exhibit pressure oscillations, loss of convergence, or complete instability.

1.7.7 Stable Finite Element Pairs

Taylor–Hood elements. Taylor–Hood elements use higher-order elements for displacement and lower-order elements for pressure:

$$V_h = [P_{k+1}]^d, \quad Q_h = P_k,$$

with $k \geq 1$. The pair $[P_2]^d - P_1$ is a standard choice and satisfies the inf–sup condition on common meshes.

MINI element. The MINI element enriches linear displacement elements with bubble functions:

$$V_h = [P_1 \oplus \text{bubble}]^d, \quad Q_h = P_1.$$

The bubble functions increase the richness of the displacement space sufficiently to satisfy the inf–sup condition.

1.7.8 Why the $P_1 - P_1$ Pair Fails: A Simple Example

Consider the equal-order approximation

$$V_h = [P_1]^d, \quad Q_h = P_1.$$

Although attractive due to its simplicity, this pair does *not* satisfy the inf–sup condition.

On a triangulated domain, there exist nonzero pressure fields $p_h \in Q_h$ that are orthogonal to the discrete divergence of all displacement fields in V_h , that is,

$$\int_{\Omega} p_h \operatorname{tr} \underline{\underline{\varepsilon}}(\underline{v}_h) dx = 0 \quad \text{for all } \underline{v}_h \in V_h.$$

Such pressure modes do not influence the displacement field and are therefore not controlled by the formulation.

As a consequence:

- the pressure is not uniquely determined,
- spurious oscillatory pressure modes may appear,
- the discrete system is unstable.

This failure is intrinsic to the choice of spaces and cannot be remedied by mesh refinement or solver techniques. It illustrates the necessity of using inf–sup stable finite element pairs.

1.7.9 Nearly Incompressible Materials

For nearly incompressible materials, the mixed formulation becomes

$$\begin{aligned} \int_{\Omega} 2\mu \underline{\varepsilon}_{\text{dev}}(\underline{u}) : \underline{\varepsilon}_{\text{dev}}(\underline{v}) \, dx - \int_{\Omega} p \, \text{tr} \, \underline{\varepsilon}(\underline{v}) \, dx &= \ell(\underline{v}), \\ \int_{\Omega} q \, \text{tr} \, \underline{\varepsilon}(\underline{u}) \, dx - \frac{1}{\kappa} \int_{\Omega} pq \, dx &= 0. \end{aligned}$$

As $\kappa \rightarrow \infty$, the incompressible formulation is recovered smoothly.

1.7.10 Summary

Incompressibility transforms elasticity into a constrained energy minimization problem. Mixed displacement–pressure formulations arise naturally from this viewpoint and lead to saddle-point systems.

Stability is governed by the inf–sup condition, which imposes compatibility requirements on displacement and pressure spaces. Stable finite element pairs such as Taylor–Hood and MINI elements satisfy this condition, while equal-order P_1 – P_1 pairs fail and lead to instability.

Mixed finite element methods are therefore essential for accurate and robust simulation of incompressible and nearly incompressible elastic materials.

1.8 Numerical Inf–Sup Test

1.8.1 Motivation

In the mixed displacement–pressure formulation introduced in Section 1.7, incompressibility is enforced through the constraint

$$\nabla \cdot \underline{u} = 0,$$

with the pressure field p acting as a Lagrange multiplier.

At the continuous level, well-posedness is ensured by the inf–sup (Ladyzhenskaya–Babuška–Brezzi) condition. At the discrete level, however, stability depends on the compatibility of the finite element spaces

$$U_h \subset U, \quad Q_h \subset L^2(\Omega).$$

Verifying analytically whether a given pair (U_h, Q_h) satisfies the inf–sup condition is generally difficult. For this reason, a numerical inf–sup test is commonly used to assess stability in practice.

1.8.2 Discrete Mixed Formulation

Recall the mixed weak problem:

Find $(\underline{u}_h, p_h) \in U_h \times Q_h$ such that

$$a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, p_h) = \ell(\underline{v}_h) \quad \forall \underline{v}_h \in U_h, \quad (1.7)$$

$$b(\underline{u}_h, q_h) = 0 \quad \forall q_h \in Q_h. \quad (1.8)$$

The elastic bilinear form is

$$a(\underline{u}, \underline{v}) = \int_{\Omega} \underline{\varepsilon}(\underline{v}) : \underline{\underline{C}} : \underline{\varepsilon}(\underline{u}) d\Omega,$$

while the displacement–pressure coupling reads

$$b(\underline{v}, q) = - \int_{\Omega} q \nabla \cdot \underline{v} d\Omega.$$

1.8.3 Discrete Inf–Sup Condition

The discrete stability requirement reads

$$\beta_h = \inf_{q_h \in Q_h} \sup_{\underline{v}_h \in U_h} \frac{b(\underline{v}_h, q_h)}{\|\underline{v}_h\|_U \|q_h\|_{L^2(\Omega)}} > 0,$$

where the displacement norm is induced by the elastic energy,

$$\|\underline{v}_h\|_U^2 = a(\underline{v}_h, \underline{v}_h).$$

The constant β_h measures the strength of the coupling between displacement and pressure spaces.

- If β_h remains bounded away from zero under mesh refinement, the discretization is stable.
- If $\beta_h \rightarrow 0$ as $h \rightarrow 0$, the inf–sup condition is violated.

1.8.4 Matrix Representation

Let

- $\mathbf{K}$: displacement stiffness matrix associated with $a(\cdot, \cdot)$,
- $\mathbf{B}$: discrete divergence matrix associated with $b(\cdot, \cdot)$,
- $\mathbf{M}_p$: pressure mass matrix in Q_h .

The algebraic mixed system reads

$$\begin{bmatrix} \mathbf{K} & \mathbf{B}^T \\ \mathbf{B} & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \mathbf{p} \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ 0 \end{Bmatrix}.$$

The discrete inf–sup constant can be expressed as

$$\beta_h^2 = \inf_{\mathbf{p} \neq 0} \frac{\mathbf{p}^T \mathbf{B} \mathbf{K}^{-1} \mathbf{B}^T \mathbf{p}}{\mathbf{p}^T \mathbf{M}_p \mathbf{p}}.$$

Defining the symmetric matrix

$$\mathbf{G} = \mathbf{B} \mathbf{K}^{-1} \mathbf{B}^T,$$

one obtains the generalized eigenvalue problem

$$\mathbf{G} \mathbf{p} = \lambda \mathbf{M}_p \mathbf{p}.$$

The smallest non-zero eigenvalue satisfies

$$\beta_h = \sqrt{\lambda_{\min}}.$$

1.8.5 Spurious Pressure Modes

Eigenvectors associated with zero eigenvalues correspond to pressure fields p_h such that

$$b(\underline{v}_h, p_h) = 0 \quad \forall \underline{v}_h \in U_h.$$

These modes lie in the kernel of $\mathbf{B}^T$ and are called *spurious pressure modes*. They lead to singular mixed systems in the incompressible limit and may produce nonphysical pressure oscillations.

1.8.6 Practical Evaluation Procedure

A typical numerical inf-sup test proceeds as follows:

1. Select a simple reference problem (e.g. unit square, plane strain elasticity).
2. Construct a sequence of meshes with characteristic size h .
3. Assemble the matrices $\mathbf{K}$, $\mathbf{B}$, and $\mathbf{M}_p$.
4. Solve the generalized eigenvalue problem

$$\mathbf{G}\mathbf{p} = \lambda\mathbf{M}_p\mathbf{p}.$$

5. Extract the smallest non-zero eigenvalue.

1.8.7 Interpretation Under Mesh Refinement

The evolution of β_h with mesh refinement determines stability:

- **Stable pair:**

$$\beta_h \rightarrow \beta_0 > 0.$$

- **Unstable pair:**

$$\beta_h \rightarrow 0.$$

In practice, one often plots

$$\log(\beta_h) \quad \text{versus} \quad \log(h),$$

where a monotonic decrease indicates violation of the inf-sup condition.

1.8.8 Connection with Volumetric Locking

Failure of the discrete inf-sup condition manifests in nearly incompressible elasticity through:

- volumetric locking,
- poor displacement convergence,
- oscillatory pressure fields.

Conversely, element pairs satisfying the inf-sup condition remain robust in the incompressible limit.

1.8.9 Inf–Sup Condition and Quasi–Incompressible Elasticity

In the quasi–incompressible formulation, the mixed problem reads:

Find $(\underline{u}_h, p_h) \in U_h \times Q_h$ such that

$$a(\underline{u}_h, \underline{v}_h) + b(\underline{v}_h, p_h) = \ell(\underline{v}_h) \quad \forall \underline{v}_h \in U_h, \quad (1.9)$$

$$b(\underline{u}_h, q_h) - \frac{1}{\kappa}(p_h, q_h) = 0 \quad \forall q_h \in Q_h, \quad (1.10)$$

where $\kappa > 0$ denotes the bulk modulus and

$$(p_h, q_h) = \int_{\Omega} p_h q_h \, d\Omega.$$

The associated algebraic system is

$$\begin{bmatrix} \mathbf{K} & \mathbf{B}^T \\ \mathbf{B} & -\frac{1}{\kappa} \mathbf{M}_p \end{bmatrix} \begin{Bmatrix} \mathbf{u} \\ \mathbf{p} \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ 0 \end{Bmatrix}.$$

For any fixed $\kappa < \infty$, this system is invertible even if the pair (U_h, Q_h) does not satisfy the discrete inf–sup condition. However, the stability of the discretization must be assessed uniformly with respect to κ .

As $\kappa \rightarrow \infty$, the block

$$-\frac{1}{\kappa} \mathbf{M}_p \longrightarrow 0,$$

and the system approaches the incompressible saddle–point problem. If the discrete inf–sup condition is not satisfied, the stability constant deteriorates as κ increases. In practice, this manifests as

- volumetric locking,
- loss of convergence for large κ ,
- spurious or oscillatory pressure fields,
- severe ill-conditioning.

Conversely, if the discrete inf–sup condition holds with a constant independent of the mesh size, the quasi–incompressible formulation remains uniformly stable as $\kappa \rightarrow \infty$. The error estimates and conditioning are then independent of the bulk modulus.

For this reason, the numerical inf–sup test is performed on the limit problem (i.e. without the $1/\kappa$ term) and provides the relevant criterion for assessing the robustness of a mixed discretization in the nearly incompressible regime.

Bibliography
